

S.R.J. Likith, Ph. D

Email jailikith@gmail.com

Phone +91 9448343837

Linkedin linkedin.com/in/likithjai

Education

May 2020 Colorado School of Mines, Golden, CO 80401, USA
Ph.D., Mechanical Engineering, Advisor: Prof. Cristian Ciobanu
THESIS - Chalcogenide-based van der Waals-Layered Materials
for Enhanced Electronic and Electromechanical Properties
GPA: 3.82

Dec 2018 Colorado School of Mines, Golden, CO 80401, USA
Master of Science, Mechanical Engineering

Jul 2014 Amrita School of Engineering, Bangalore, India
Bachelor of Technology, Mechanical Engineering

Employment

2022-2023 Deloitte, Lake Mary, FL 32801, USA
Data Analyst

- Analysed **big data systems** and designed **custom data solutions** for the **S&P 500** component bank holding company as a client: **M&T Bank** using advanced technologies such as Snowflake, Hadoop, SQL, Python, and Databricks
- Led a cross-disciplinary team including data engineers, analysts, and product owners in efforts to overhaul and debug the **Mortgage Data** system of records, securing a **Mortgage Data Superstar** award from the client
- Took charge of merging of a **commercial equipment financing** system of records of the client and an acquired bank: **People's United Bank (PUB)** including around 200 profitability, regulatory, and FDIC reporting requirements. This involved tracking the relevant financial data elements of the two entities and tallying all the General Ledger (GL) entries of the fee and revenue streams, resulting in **one cohesive system of records on the Snowflake platform**.
- Client-facing and back-end facing communication to test and implement solutions end-to-end.

2020-2021 University of Florida, Gainesville, FL 32608, USA
Postdoctoral Researcher

- Data Analysis, website content, and database management** for the Engaged Quality Instruction through Professional Development (EQUIPD) group under the advisership of Dr. Nancy Ruzycki (Department of Materials Science and Engineering)
- Using analytical and programming skills (**Python**) to interpret and investigate large datasets. This was used to drive decision-making for the group and to understand the experiences of the teachers.
- Written content for the group's website
- SQL-based inventory database management**

2021-2022 Colorado School of Mines, Golden, CO 80401, USA
Postdoctoral Researcher

- Using Density Functional Theory (DFT) to analyse the dynamical stability of Ti-based alloy systems for the Center for Advanced Non-Ferrous Structural Alloys (CANFSA) group under the advisership of Dr. Amy Clarke (Department of Metallurgical and Materials Engineering)
- Leveraging remote **High Performance Computing (HPC) clusters** to investigate massive datasets and **data visualization**

Teaching

2015-2016, Colorado School of Mines, Golden, CO 80401, USA

2019-2020 Teaching Assistant

- Courses: Solid Mechanics, Advanced Solid Mechanics, Dynamics
 - Conducted recitations, helped students with understanding various aspects of the course and problem-solving, and graded weekly assignments
 - Planning content for lectures, managing co-TAs, and various pedagogy strategies
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Publications, Presentations, and Posters

Publications

- **S.R.J. Likith**, C.A. Farebrow, S. Manna, A. Abdulslam, V. Stevanović, D.A. Ruddy, J.A. Schaidle, D.J. Robichaud, C.V. Ciobanu
Thermodynamic Stability of Molybdenum Oxycarbides Formed from Orthorhombic Mo₂C in Oxygen-Rich Environments
Journal of Physical Chemistry C, 122, 1223 (2018)
- G. Stan, C.V. Ciobanu, **S.R.J. Likith**, A. Rani, S. Krylyuk, A.V. Davydov
Doping of MoTe₂ via surface charge-transfer in air
ACS Applied Materials and Interfaces 12 (15), 18182-18193
- **S.R.J. Likith**, G.L. Brennecke, C.V. Ciobanu
Interlayer Registry Effects on the Electronic and Piezoelectric Properties of Transition Metal Dichalcogenide Bilayer Heterostructures
Journal of Vacuum Science & Technology A: Vacuum, Surfaces, and Films (Submitted)
- **S.R.J. Likith** and C.V. Ciobanu Electronic and Electromechanical Properties of Vertical and Lateral 2D Heterostructures
Modeling, Characterization and Production of Nanomaterials, 2nd Edition (Book chapter)
- **S.R.J. Likith**, C.V. Ciobanu
Structure and Stability of van der Waals-Layered Group-IV Monochalcogenides
Journal of Vacuum Science & Technology A: Vacuum, Surfaces, and Films 40, 5, 052202 (2022)
- C.B. Finfrock, B. Ellyson, **S.R.J. Likith**, D. Smith, C.J. Rietema, A.J. Saville, M.M. Thrun, C.G. Becker, A.L. Arujo, E.J. Pavlina, J. Hu, J.S. Park, A.J. Clarke, K.D. Clarke Elucidating the Temperature Dependence of TRIP in Q&P Steels using Synchrotron X-Ray Diffraction, Constituent Phase Properties, and Strain-based Kinetics Models
Acta Materialia 237, 118126 (2022)

Conference Proceedings

- **S.R.J. Likith, (Designated Speaker)** Amy Clarke
Local Distortion Effects on the Dynamic Lattice Stability in the BCC Phase of Titanium and its Alloys
TMS Annual Meeting & Exhibition 2022
- G. Stan, C.V. Ciobanu, **S.R.J. Likith**, A. Rani, S. Krylyuk, A.V. Davydov
Native and induced surface charge-transfer doping of MoTe₂
Bulletin of the American Physical Society 2020
- C.A. Farberow, **S.R.J. Likith**, S. Manna, A. Abdulslam, V. Stevanović, D.A. Ruddy, J.A. Schaidle, D.J. Robichaud, C.V. Ciobanu
Thermodynamic Stability of Molybdenum Oxycarbides Formed from Orthorhombic Mo₂C in Oxygen-Rich Environments
North American Catalysis Society Meeting 2019

Posters

- **S.R.J. Likith**, and C.V. Ciobanu
Quasi-2D Group-IV Monochalcogenides: A DFT Search
Materials Research Society Dec 2019
 - **S.R.J. Likith**, and C.V. Ciobanu
Performance of van der Waals Density Functionals for Group-IV Monochalcogenides
Materials Research Society Dec 2019
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Research Projects

- **2D Materials**
Colorado School of Mines, Golden, CO 80401, USA
 - Studied the electronic and piezoelectric properties of 2D and quasi-2D (van der Waals-layered) materials using atomistic modeling (Density Functional Theory, DFT)
 - Modeling van der Waals-dominated systems, computational discovery strategies, and automating tasks
 - Text file manipulation, compiling relevant information from large numbers of files, and post-processing data (e.g., plotting) using Python
 - **Piezoelectric materials**
Colorado School of Mines, Golden, CO 80401, USA
 - Computational discovery of Perovskite Nitrides and Transition Metal Dichalcogenide heterostructures for piezoelectric applications
 - Computational discovery of Perovskite Nitrides and Transition Metal Dichalcogenide heterostructures for piezoelectric applications
 - Interplay between electronic and mechanical behaviour of materials
 - Analysing large sets of data and mechanical behaviour of materials
 - **Transition metal carbide catalysts**
Colorado School of Mines, Golden, CO 80401, USA
 - Studied the structure and composition of metal carbide catalysts for ex-situ catalytic fast pyrolysis
 - High-throughput computation strategies, thermodynamics and energetics of adsorption
 - Large-scale text file manipulation using bash, and post-processing data using MATLAB
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Skills

■ Computer Skills

Operating Systems: Linux, macOS, Windows

Languages: Python, MATLAB, bash

Applications: Vienna *Ab-initio* Simulation Package (VASP), Visualization for Electronic and STructural Analysis (VESTA), Amsterdam Density Functional (ADF), Visual Molecular Dynamics (VMD), Jmol, p4vasp

Word-processing: LaTeX, Emacs, MS Office, LibreOffice/OpenOffice

■ Communication Skills

Multilingual, presentation, teaching (private tutoring, lecturing), technical writing

Outreach and other Activities

- 2017-2020** Colorado High-School Science Outreach (Golden area middle and high-schools, CO, USA)
Outreach Speaker
- Presented research-related science topics to middle and high-school students
 - Pedagogy and presentation skills
 - Ability to explain a complex concept to someone outside the field or is at a middle or high-school level
- 2019** Science Olympiad (Greenwood Village, CO, USA)
Colorado State-level Event Supervisor
- Supervised a chemistry experiment event for high-school students from around the state
 - Explained some basic chemistry concepts on-the-fly, ensured safety of all involved
 - Hands-on teaching and managing event logistics
- 2019** STEM Guitar Workshop (Colorado School of Mines, Golden, CO 80401, USA)
Workshop Participant
- Built a fully functional and tuned electric guitar
 - Wood-working, dip-painting, electronic circuitry, soldering, intonation and tuning
 - Applied materials science, physics, and electronics concepts to the building and functioning of an electric guitar
- 2019** What is Data Science?
Online course designed by IBM (Coursera)
- Studied the meaning, importance, and power of Data Science in various fields
 - Learnt the structure, and important parts of a (data science) report meant for non-science/management peers and superiors
 - Role of big data, data-mining, statistics, and regression in Data Science
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Interests

Guitar-playing, biomimetics, pedagogy, cooking, metal and post-rock music, traveling